oscillation: it did not change when the chandelier slowed and oscillated with diminished amplitude. The oscillations all had the same duration.

It's a fine story but, on reflection, it leaves us perplexed – and this perplexity goes to the heart of the problem of time. How could Galileo know that *his own individual pulse-beats* all lasted for the same amount of time?[fn41]

Not many years after Galileo, doctors began to measure their patients' pulses by using a watch – which is nothing, after all, but a pendulum. So we use the beats to assure ourselves that the pendulum is regular, and then the pendulum to ascertain the regularity of the pulse-beats. Is this not somewhat circular? What does it mean?

It means that we, in reality, never measure time itself; we always measure the physical variables *A*, *B*, *C* … (oscillations, beats, and many other things) and compare one variable with another, that is to say, we measure the functions *A(B)*, *B(C)*, *C(A)*, and so on. We can count how many beats for each oscillation; how many oscillations for every tick of my stopwatch; how many ticks of my stopwatch between intervals of the clock on the bell-tower …

The point is that it is *useful* to imagine that a variable *t* exists – the 'true time' – which underpins all those movements, even if we cannot measure it directly. We write the equations for the physical variables with regard to this *unobservable t*, equations which tell us how things change in *t*; that is, for instance, how much time it takes for each oscillation, and how long each heart-beat lasts. From this, we can derive how the variables change in relation to each other – how many heartbeats there are in one oscillation – and compare this prediction with what we observe in the world. If the predictions are correct, we trust that this complicated schema is a sound one and, in particular, that it is useful to employ the variable of time *t*, even if we cannot measure it directly.

In other words, the existence of the variable time is a useful assumption, not the result of an observation.

It was Newton who understood all of this: he understood that this was a good way to proceed, and clarified and developed this schema. Newton asserts explicitly in his book that we can't ever measure the true time *t* but, if we *assume* that it exists, we can set up an efficient framework to describe nature.

Having clarified this, we can return to quantum gravity and the meaning of the statement that 'time does not exist'. It simply means that the Newtonian schema no longer works when we are dealing with small things. It was a good one, but only for large things.

If we want to understand the world widely, if we want to understand how it functions in the less familiar situations where quantum gravity matters, we need to abandon this schema. The idea of a time *t* which flows by itself, and in relation to which all things evolve, is no longer a useful one. The world is